

Summary of the Customers.csv Data Analysis

1. Introduction

The customers.csv dataset was analyzed to understand customer purchasing behavior across different product categories, customer regions, and sales channels. The analysis used Python-based data analysis techniques to explore the structure of the dataset, summarize numerical variables, examine categorical distributions, identify relationships between spending categories, and visualize important patterns.

The dataset contains 440 customer records and 8 columns, with no missing values, providing a complete dataset for exploratory analysis.

2. Data Inspection and Preparation

The first stage involved inspecting the dataset to understand its size, variables, data types, and overall structure. Missing-value checks were performed to determine whether the dataset required substantial cleaning. The analysis found that the dataset contained 440 observations and 8 columns, with zero missing values. This meant that the data could be used directly for exploratory analysis without requiring extensive missing-value treatment.

3. Categorical Data Analysis

The categorical variables were examined to understand how customers were distributed across sales channels and geographical regions.

Channel 1 contained 298 customers, while Channel 2 contained 142 customers. Region 1 contained 77 customers, Region 2 contained 47 customers, and Region 3 contained 316 customers. This shows that Region 3 represents the largest proportion of customers, while Channel 1 contains more customers than Channel 2.

4. Numerical Data Analysis

The numerical variables represented customer spending in six product categories: Fresh, Milk, Grocery, Frozen, Detergents/Paper, and Delicatessen. Descriptive statistics were calculated for each category, including the mean, standard deviation, minimum, quartiles, median, and maximum.

Fresh had the highest average spending at approximately 12,000.3, followed by Grocery at approximately 7,951.3 and Milk at approximately 5,796.3. Delicatessen had the lowest average spending at approximately 1,524.9. The large differences between minimum and maximum values indicate considerable variation in customer spending.

5. Distribution Analysis

Histograms and other distribution visualizations were used to understand how customer spending was distributed. The spending variables showed strong right-skewed distributions, meaning that most customers recorded relatively lower or moderate spending while a smaller number recorded exceptionally high spending. These high-value customers appear as potential outliers in several categories. Box plots were useful for identifying the spread of spending and highlighting unusually large customer purchases.

6. Channel-Based Comparison

Customer spending was compared between the two sales channels. The analysis showed that Channel 2 had substantially higher average spending on Milk, Grocery, and Detergents/Paper than Channel 1. This indicates that purchasing behavior differs considerably between the two channels and provides useful insight into how customers using different channels behave.

7. Correlation Analysis

A correlation analysis was performed to examine relationships between the different product-spending variables. One of the strongest relationships identified was between Grocery and Detergents/Paper, with a correlation of approximately 0.92. This represents a very strong positive relationship, meaning that customers who spend more on Grocery products also tend to spend more on Detergents/Paper products.

Correlation indicates association rather than proving that one type of spending causes another.

8. Data Visualization

Several visualization techniques were used to make the analysis easier to understand. These included bar charts for categorical comparisons, histograms for numerical distributions, box plots for identifying spread and potential outliers, correlation heatmaps for examining relationships between numerical variables, and comparative visualizations for analyzing spending across channels and categories.

9. Major Findings

1. The dataset contains 440 customers and 8 variables with no missing values.
2. Channel 1 contains more customers than Channel 2.
3. Region 3 contains the largest number of customers.
4. Fresh products have the highest average spending among the six product categories.
5. Delicatessen has the lowest average spending.
6. Channel 2 customers spend substantially more on Milk, Grocery, and Detergents/Paper than Channel 1 customers.
7. Grocery and Detergents/Paper have a very strong positive correlation of approximately 0.92.
8. Customer spending distributions are strongly right-skewed.
9. Several categories contain high-value customers whose spending is considerably higher than that of the typical customer.

10. Business Value of the Analysis

The analysis provides useful information for understanding customer purchasing behavior. Businesses can use these findings to identify high-value customers, understand differences between sales channels, examine product combinations, and improve customer segmentation.

The strong relationship between Grocery and Detergents/Paper spending may be useful when considering product bundling, cross-selling, promotions, or targeted marketing strategies. Differences between Channel 1 and Channel 2 also suggest that customers using different channels may require different marketing approaches and product strategies.

11. Limitations

The analysis is primarily descriptive and exploratory. It identifies patterns and relationships within the available customer data but does not establish causal relationships. The presence of strong correlations does not necessarily mean that one product category causes customers to purchase another. Additional

statistical testing or predictive modeling would be required to investigate these relationships more deeply.

12. Overall Conclusion

Overall, the analysis of customers.csv successfully transformed raw customer records into meaningful information about purchasing behavior. The work involved data inspection, data-quality checking, descriptive statistics, categorical analysis, numerical analysis, distribution analysis, channel comparisons, correlation analysis, and data visualization.

The results show substantial variation in customer spending and reveal meaningful differences across sales channels and product categories. In particular, the strong relationship between Grocery and Detergents/Paper spending provides an important insight into customer purchasing patterns.

The completed analysis can serve as a foundation for more advanced customer analytics, including customer segmentation, purchasing-behavior prediction, market-basket analysis, targeted marketing, customer-value analysis, and recommendation systems.